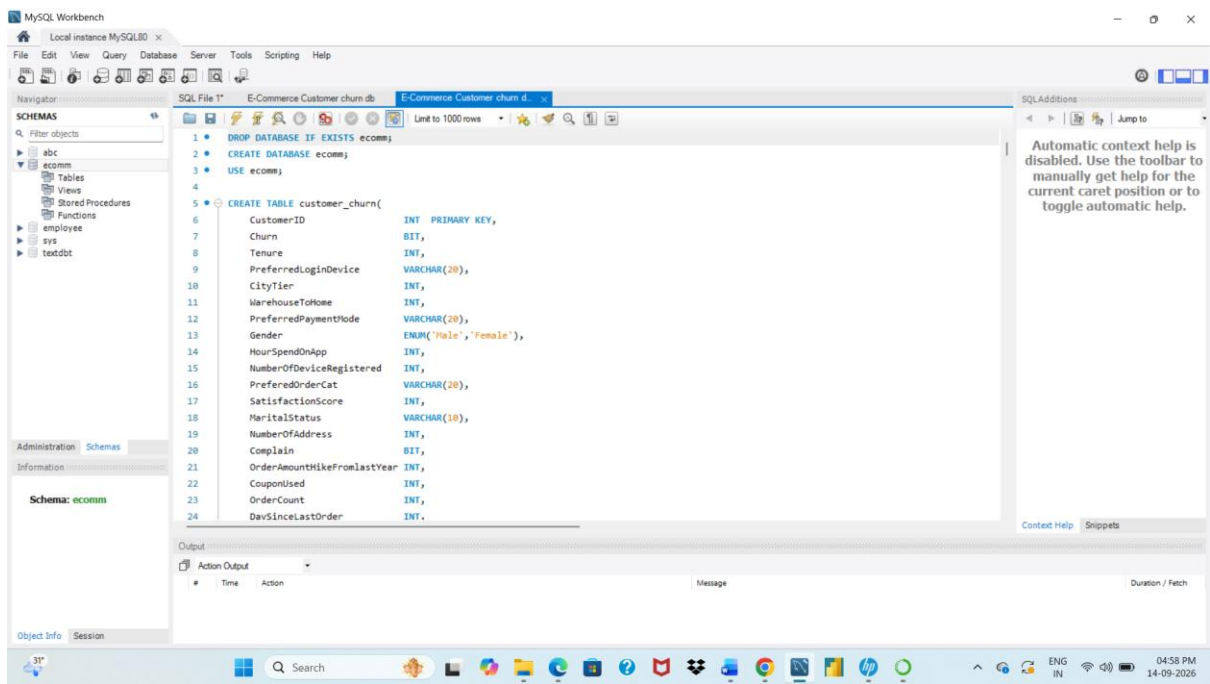
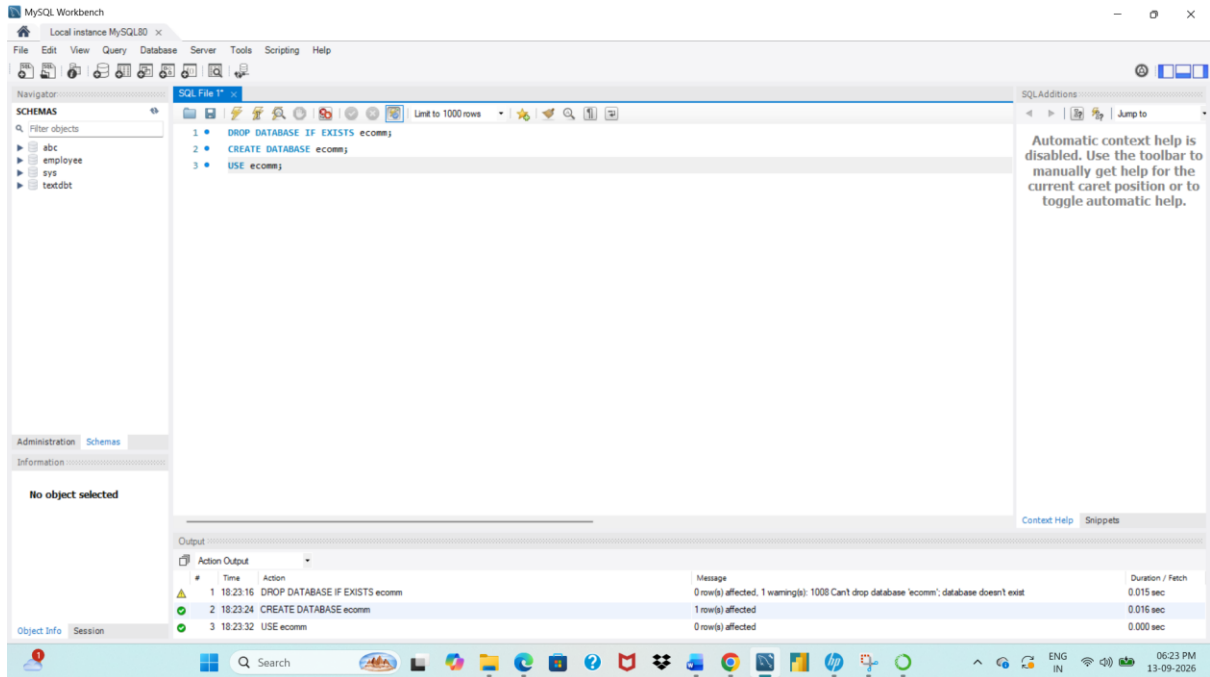


Module End Assignment 2 MySQL E-Commerce Customer Churn Analysis



MySQL Workbench

Local instance MySQL80 x

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Information

Schema: ecomm

SQL File 1*

E-Commerce Customer churn db

E-Commerce Customer churn d...

Limit to 1000 rows

SQL Editor

```

6558 (55630,8,8,'Mobile Phone',1,15,'Credit Card','Male',3,2,'Laptop & Accessory',3,'Married',4,8,13,2,3,169);
6559
6560 SHOW TABLES;
6561 SELECT * FROM customer_churn;
6562

```

Result Grid

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDevicesRegistered	PreferredOrder
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Access
50002	1	8	Phone	1	8	UPI	Male	3	4	Mobile
50003	1	30	Phone	1	30	Debit Card	Male	2	4	Mobile
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Access
50005	1	0	Phone	1	12	CC	Male	3	3	Mobile
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone
50007	1	3	Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Access
50008	1	6	Phone	1	6	CC	Male	3	3	Mobile
50009	1	13	Phone	3	9	E wallet	Male	3	4	Mobile
50010	1	31	Phone	1	31	Debit Card	Male	2	5	Mobile
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion
50013	1	0	Phone	1	11	COD	Male	2	3	Mobile
50014	1	0	Phone	1	15	CC	Male	3	4	Mobile
50015	1	9	Mobile Phone	3	15	Credit Card	Male	3	4	Fashion
50016	1	2	Phone	2	12	UPI	Male	3	3	Mobile
50017	1	0	Computer	1	12	Debit Card	Female	4	4	Mobile
50018	1	0	Mobile Phone	3	11	E wallet	Male	2	4	Laptop & Access

customer_churn 2 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	17:15:09	SHOW TABLES	1 row(s) returned	0.000 sec / 0.000 sec
9	17:15:54	SELECT * FROM ecomm LIMIT 0, 1000	Error Code: 1146: Table 'ecomm.eccomm' doesn't exist	0.000 sec
10	17:16:20	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
11	17:16:21	SELECT COUNT(*) AS Total_Customers FROM customer_churn	1 row(s) returned	0.078 sec / 0.000 sec

Object Info Session

31°

Search

ENG IN

05:17 PM 14-09-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

○ Total Number of Customer

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

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Administration Schemas

Information

Schema: ecomm

SQL File 1*

E-Commerce Customer churn db

E-Commerce Customer churn d...

Limit to 1000 rows

SQL Editor

```

6552 (55624,8,5,'Computer',1,12,'Credit Card','Male',4,4,'Laptop & Accessory',5,'Single',2,8,28,2,2,MULTI,224),
6553 (55625,8,1,'Mobile Phone',3,12,'UPI','Female',2,5,'Mobile Phone',3,'Single',2,8,19,2,2,1,155),
6554 (55626,8,18,'Computer',1,30,'Credit Card','Male',3,2,'Laptop & Accessory',1,'Married',6,8,18,1,2,4,151),
6555 (55627,8,13,'Mobile Phone',1,13,'Credit Card','Male',3,5,'Fashion',5,'Married',6,8,16,1,2,MULTI,225),
6556 (55628,8,1,'Mobile Phone',1,11,'Debit Card','Male',3,2,'Laptop & Accessory',4,'Married',3,1,21,1,2,4,186),
6557 (55629,8,23,'Computer',3,9,'Credit Card','Male',4,5,'Laptop & Accessory',4,'Married',4,8,15,2,2,9,179),
6558 (55630,8,8,'Mobile Phone',1,15,'Credit Card','Male',3,2,'Laptop & Accessory',3,'Married',4,8,13,2,2,3,169);
6559
6560 SHOW TABLES;
6561 SELECT * FROM customer_churn;
6562
6563 SELECT COUNT(*) AS Total_Customers FROM customer_churn;
6564
6565
6566

```

Result Grid

Total_Customers
5630

Result 3 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	17:15:54	SELECT * FROM ecomm LIMIT 0, 1000	Error Code: 1146: Table 'ecomm.eccomm' doesn't exist	0.000 sec
10	17:16:20	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
11	17:53:21	SELECT COUNT(*) AS Total_Customers FROM customer_churn LIMIT 0, 1000	1 row(s) returned	0.078 sec / 0.000 sec

Object Info Session

Search

ENG IN

05:34 PM 14-09-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

○ Customer Churn Table schema

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'ecom' database selected. The main window shows the 'E-Commerce Customer churn db' database with the 'customer_churn' table selected. The 'Result Grid' displays the table schema with columns: CustomerID, Churn, Tenure, PreferredLoginDevice, CityTier, WarehouseToHome, PreferredPaymentMode, Gender, HourSpendOnApp, NumberOfDeviceRegistered, PreferredOrderCat, SatisfactionScore, MaritalStatus, NumberOfAddress, and Cmmnln. The 'Output' pane shows the execution of a query: 'SELECT * FROM customer_churn LIMIT 0, 1000'. The 'Message' column indicates '1000 row(s) returned'.

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PK		
Churn	bit(1)	YES			
Tenure	int	YES			
PreferredLoginDevice	varchar(20)	YES			
CityTier	int	YES			
WarehouseToHome	int	YES			
PreferredPaymentMode	varchar(20)	YES			
Gender	enum('Male','Female')	YES			
HourSpendOnApp	int	YES			
NumberOfDeviceRegistered	int	YES			
PreferredOrderCat	varchar(20)	YES			
SatisfactionScore	int	YES			
MaritalStatus	varchar(10)	YES			
NumberOfAddress	int	YES			
Cmmnln	bit(1)	YES			

○ Checking Missing Values

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'ecom' database selected. The main window shows the 'E-Commerce Customer churn db' database with the 'customer_churn' table selected. The 'Result Grid' displays the query results for the 'customer_churn' table. The 'Output' pane shows the execution of a query: 'SELECT COUNT(*) AS Total_Rows, COUNT(WarehouseToHome) AS WarehouseToHome_Count, COUNT(HourSpendOnApp) AS HourSpendOnApp_Count, COUNT(PreferredPaymentMode) AS PreferredPaymentMode_Count, COUNT(Gender) AS Gender_Count, COUNT(HourSpendOnApp) AS HourSpendOnApp_Count, COUNT(NumberOfDeviceRegistered) AS NumberOfDeviceRegistered_Count, COUNT(PreferredOrderCat) AS PreferredOrderCat_Count, COUNT(SatisfactionScore) AS SatisfactionScore_Count, COUNT(MaritalStatus) AS MaritalStatus_Count, COUNT(NumberOfAddress) AS NumberOfAddress_Count, COUNT(Cmmnln) AS Cmmnln_Count FROM customer_churn;'. The 'Message' column indicates '1 row(s) returned'.

Field	Type	Null	Key	Default	Extra
Total_Rows	int	NO			
WarehouseToHome_Count	int	YES			
HourSpendOnApp_Count	int	YES			
PreferredPaymentMode_Count	int	YES			
Gender_Count	int	YES			
HourSpendOnApp_Count	int	YES			
NumberOfDeviceRegistered_Count	int	YES			
PreferredOrderCat_Count	int	YES			
SatisfactionScore_Count	int	YES			
MaritalStatus_Count	int	YES			
NumberOfAddress_Count	int	YES			
Cmmnln_Count	int	YES			

Data Cleaning

Handling Missing Values and Outliers:

- Impute mean for the following columns, and round off to the nearest integer if required

Impute mean WarehouseToHome

The screenshot shows the MySQL Workbench interface with the following SQL queries:

```
5670 COUNT(WarehouseToHome) AS WarehouseToHome_NotNull,
5671 COUNT(HourSpendOnApp) AS HourSpendOnApp_NotNull,
5672 COUNT(OrdersAmountHikeFromLastYear) AS OrdersAmountHike_NotNull,
5673 COUNT(DaysSinceLastOrder) AS DaysSinceLastOrder_NotNull,
5674 COUNT(Tenure) AS Tenure_NotNull,
5675 COUNT(CouponUsed) AS CouponUsed_NotNull,
5676 COUNT(OrdersCount) AS OrdersCount_NotNull FROM customer_churn;
5677
5678
5679 (SELECT ROUND(AVG(WarehouseToHome))
5680 FROM (SELECT WarehouseToHome
5681 FROM customer_churn
5682 WHERE WarehouseToHome IS NOT NULL) as temp );
5683
5684
5685 UPDATE ecommerce_customer_churn
```

The Result Grid shows the output of the first query:

ROUND(AVG(WarehouseToHome))
16

The Output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
19	19:18:50	(SELECT ROUND(AVG(WarehouseToHome)) FROM (SELECT WarehouseToHome FROM ecommerce_cust...	Error Code: 1146. Table 'ecommerce_customer_churn' doesn't exist	0.031 sec
20	19:19:28	(SELECT ROUND(AVG(WarehouseToHome)) FROM (SELECT WarehouseToHome FROM customer_churn ...	Error Code: 1248. Every derived table must have its own alias	0.031 sec
21	19:19:41	(SELECT ROUND(AVG(WarehouseToHome)) FROM (SELECT WarehouseToHome FROM customer_churn ...	1 row(s) returned	0.000 sec / 0.000 sec

The screenshot shows the MySQL Workbench interface with the following SQL queries:

```
5686 UPDATE customer_churn
5687 SET WarehouseToHome = (SELECT ROUND(AVG(WarehouseToHome))
5688 FROM (SELECT WarehouseToHome
5689 FROM customer_churn
5690 WHERE WarehouseToHome IS NOT NULL) AS temp)
5691 WHERE WarehouseToHome IS NULL;
5692
5693
5694
5695
```

The Result Grid shows the output of the first query:

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDevicesRegistered	PreferredOrder
90997	0	9	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Access
90998	0	2	Phone	1	9	Credit Card	Male	2	3	Mobile Phone
90999	0	8	Mobile Phone	1	16	Credit Card	Female	2	3	Laptop & Access
91000	0	12	Mobile Phone	3	12	Credit Card	Male	2	1	Fashion

The Output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
25	19:24:34	SET SQL_SAFE_UPDATES = 0	0 row(s) affected	0.016 sec
26	19:24:52	UPDATE customer_churn SET WarehouseToHome = (SELECT ROUND(AVG(WarehouseToHome)) FROM (S...	251 row(s) affected Rows matched: 251 Changed: 251 Warnings: 0	0.047 sec
27	19:25:11	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

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customer_churn

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employee

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testdbt

SQL File 1* E-Commerce Customer churn db E-Commerce Customer churn d...

Limit to 1000 rows

SQL Editor

```
5668
5669 SELECT COUNT(*) AS Total_Rows,
5670 COUNT(WarehouseToHome) AS WarehouseToHome_NotNull,
5671 COUNT(HourSpentOnApp) AS HourSpentOnApp_NotNull,
5672 COUNT(OrdAmountHikeFromLastYear) AS OrdAmountHike_NotNull,
5673 COUNT(DaySinceLastOrder) AS DaySinceLastOrder_NotNull,
5674 COUNT(Tenure) AS Tenure_NotNull,
5675 COUNT(CouponUsed) AS CouponUsed_NotNull,
5676 COUNT(OrdCount) AS OrdCount_NotNull FROM customer_churn;
5677
5678
5679 (SELECT ROUND(AVG(WarehouseToHome))
5680 FROM (SELECT WarehouseToHome
5681 FROM customer_churn
5682 WHERE WarehouseToHome IS NOT NULL) as temp );
5683
```

Result Grid

Total_Rows	WarehouseToHome_NotNull	HourSpentOnApp_NotNull	OrdAmountHike_NotNull	DaySinceLastOrder_NotNull	Tenure_NotNull	CouponUsed_NotNull	OrdCount_NotNull
5630	5630	5375	5365	5323	5366	5374	5372

Schema: ecomm

Output

Action Output

#	Time	Action	Message	Duration / Fetch
26	19:24:52	UPDATE customer_churn SET WarehouseToHome = (SELECT ROUND(AVG(WarehouseToHome)) FROM (SELECT WarehouseToHome FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
27	19:25:11	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
28	19:28:03	SELECT COUNT(*) AS Total_Rows, COUNT(WarehouseToHome) AS WarehouseToHome_NotNull, COUNT...	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

29°

Search

ENG IN

07:28 PM 14-09-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

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SQL File 1* E-Commerce Customer churn db E-Commerce Customer churn d...

Limit to 1000 rows

SQL Editor

```
49 (59021,1,0,'Mobile Phone',3,12,'Debit Card','Male',3,5,'Laptop & Accessory',3,'Divorced',5,1,18,6,7,162),
50 (59022,1,20,'Mobile Phone',1,20,'Credit Card','Female',3,3,'Fashion',2,'Divorced',2,0,12,11,15,6,203),
51 (59023,1,0,'Mobile Phone',3,28,'E wallet','Male',2,3,'Mobile Phone',3,'Single',2,1,19,0,1,0,117),
52 (59024,1,0,'Phone',3,26,'Debit Card','Female',3,5,'Laptop & Accessory',3,'Divorced',4,1,11,1,2,146),
53 (59025,1,14,'Computer',1,14,'Debit Card','Male',2,5,'Fashion',2,'Single',3,0,15,0,1,3,207),
54 (59026,1,0,'Phone',1,15,'Credit Card','Female',2,3,'Mobile Phone',5,'Single',0,0,22,1,4,143),
55 (59027,0,6,'Mobile Phone',3,6,'E wallet','Male',3,3,'Fashion',4,'Divorced',2,0,13,1,6,173),
56 (59028,0,0,'Phone',3,12,'E wallet','Male',2,3,'Laptop & Accessory',3,'Divorced',3,1,20,0,4,5,123),
57 (59029,0,18,'Phone',1,0,'Debit Card','Male',2,3,'Laptop & Accessory',4,'Married',0,0,18,1,1,15,123),
58 (59030,0,5,'Computer',3,14,'E wallet','Female',2,3,'Fashion',2,'Single',2,0,14,2,3,7,190),
59 (59031,0,2,'Computer',1,0,'CC','Male',2,3,'Laptop & Accessory',3,'Divorced',2,0,13,0,1,0,143),
60 (59032,0,0,'Mobile Phone',1,13,'Credit Card','Male',2,4,'Laptop & Accessory',3,'Divorced',3,0,17,1,1,0,161),
61 (59033,0,30,'Phone',1,15,'CC','Female',3,4,'Mobile',4,'Divorced',5,1,20,1,1,0,133),
62 (59034,0,13,'Mobile Phone',3,10,'E wallet','Male',3,4,'Fashion',2,'Divorced',0,0,11,2,2,11,209),
63 (59035,0,0,'Computer',3,0,'E wallet','Female',3,3,'Mobile',3,'Divorced',7,0,18,1,1,2,122),
64 (59036,0,0,'Phone',1,0,'CC','Male',3,3,'Mobile',2,'Divorced',2,0,24,0,2,2,123),
```

Result Grid

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpentOnApp	NumberOfDeviceRegistered	PreferredOrder
50035	0	0	Computer	3	16	E wallet	Female	3	3	Mobile
50036	0	0	Phone	1	15	CC	Male	3	3	Mobile
50037	0	0	Computer	1	15	CC	Male	2	3	Mobile
50038	0	1	Phone	1	27	CC	Male	3	3	Mobile

Schema: ecomm

Output

Action Output

#	Time	Action	Message	Duration / Fetch
27	19:25:11	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
28	19:28:03	SELECT COUNT(*) AS Total_Rows, COUNT(WarehouseToHome) AS WarehouseToHome_NotNull, COUNT...	1 row(s) returned	0.000 sec / 0.000 sec
29	12:45:51	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.047 sec / 0.000 sec

Object Info Session

29°

Search

ENG IN

12:48 PM 15-09-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Impute mean HourSpendOnApp

The screenshot shows the MySQL Workbench interface with the 'E-Commerce Customer churn db' database selected. The SQL editor contains the following queries:

```
5687 FROM (SELECT warehouseToHome
5688 FROM customer_churn
5689 WHERE warehouseToHome IS NOT NULL) AS temp)
5690 WHERE warehouseToHome IS NULL;
5691
5692 • UPDATE customer_churn
5693 SET HourSpendOnApp = (SELECT ROUND(AVG(HourSpendOnApp))
5694 FROM (SELECT HourSpendOnApp
5695 FROM customer_churn
5696 WHERE HourSpendOnApp IS NOT NULL) AS temp)
5697 WHERE HourSpendOnApp IS NULL;
5698
5699
5700
5701
5702
5703
5704
5705
5706
5707
5708
5709
5710
5711
5712
```

The Output pane shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
28	19:28:03	SELECT COUNT(*) AS Total_Rows, COUNT(WarehouseToHome) AS WarehouseToHome_NotNull, COUNT...	1 row(s) returned	0.000 sec / 0.000 sec
29	12:45:51	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.047 sec / 0.000 sec
30	12:51:21	UPDATE customer_churn SET HourSpendOnApp = (SELECT ROUND(AVG(HourSpendOnApp)) FROM (SEL...	255 row(s) affected Rows matched: 255 Changed: 255 Warnings: 0	0.063 sec

The screenshot shows the MySQL Workbench interface with the 'E-Commerce Customer churn db' database selected. The SQL editor contains the following queries:

```
5663
5664 • SELECT COUNT(*) AS Total_Customers
5665 FROM customer_churn;
5666
5667 • DESCRIBE customer_churn;
5668
5669 • SELECT COUNT(*) AS Total_Rows,
5670 COUNT(warehouseToHome) AS warehouseToHome_NotNull,
5671 COUNT(HourSpendOnApp) AS HourSpendOnApp_NotNull,
5672 COUNT(OrderByAmountBikeFromLastYear) AS OrderAmountBike_NotNull,
5673 COUNT(DaysSinceLastOrder) AS DaysSinceLastOrder_NotNull,
5674 COUNT(Tenure) AS Tenure_NotNull,
5675 COUNT(CouponUsed) AS CouponUsed_NotNull,
5676 COUNT(OrderByCount) AS OrderCount_NotNull FROM customer_churn;
5677
5678
5679 • (SELECT ROUND(AVG(warehouseToHome))
5680 FROM (SELECT warehouseToHome
```

The Result Grid shows the output of the queries:

Total_Rows	WarehouseToHome_NotNull	HourSpendOnApp_NotNull	OrderAmountBike_NotNull	DaysSinceLastOrder_NotNull	Tenure_NotNull	CouponUsed_NotNull	OrderCount_NotNull
5630	5630	5630	5365	5323	5366	5374	5372

The Output pane shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
30	12:51:21	UPDATE customer_churn SET HourSpendOnApp = (SELECT ROUND(AVG(HourSpendOnApp)) FROM (SEL...	255 row(s) affected Rows matched: 255 Changed: 255 Warnings: 0	0.063 sec
31	12:51:58	UPDATE customer_churn SET HourSpendOnApp = (SELECT ROUND(AVG(HourSpendOnApp)) FROM (SEL...	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0	0.000 sec
32	12:52:04	SELECT COUNT(*) AS Total_Rows, COUNT(WarehouseToHome) AS WarehouseToHome_NotNull, COUNT...	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL80 x

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ecom

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testdbt

SQL File 1* E-Commerce Customer churn db E-Commerce Customer churn d...

Limit to 1000 rows

5681 FROM customer_churn

5682 WHERE WarehouseToHome IS NOT NULL) AS temp);

5683

5684

5685 • UPDATE customer_churn

5686 SET WarehouseToHome = (SELECT ROUND(AVG(WarehouseToHome))

5687 FROM (SELECT WarehouseToHome

5688 FROM customer_churn

5689 WHERE WarehouseToHome IS NOT NULL) AS temp)

5690 WHERE WarehouseToHome IS NULL;

5691

5692 • SELECT ROUND(AVG(HourSpendOnApp))

5693 FROM (SELECT HourSpendOnApp

5694 FROM customer_churn

5695 WHERE HourSpendOnApp IS NOT NULL) AS temp);

5696

5697 • UPDATE customer_churn

5698 SET HourSpendOnApp = (SELECT ROUND(AVG(HourSpendOnApp))

Result Grid

Filter Rows

Exports

Wrap Cell Contents

Read Only

Context Help

Snippets

Schema: ecomm

Information

Object Info

Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
32	12:52:04	SELECT COUNT(*) AS Total_Rows, COUNT(WarehouseToHome) AS WarehouseToHome_NotNull, COUNT...	1 row(s) returned	0.000 sec / 0.000 sec
33	12:53:24	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.015 sec
34	12:59:12	(SELECT ROUND(AVG(HourSpendOnApp)) FROM (SELECT HourSpendOnApp FROM customer_churn WH...	1 row(s) returned	0.000 sec / 0.000 sec

12:59 PM 15-09-2026

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

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ecom

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customer_churn

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sys

testdbt

SQL File 1* E-Commerce Customer churn db E-Commerce Customer churn d...

Limit to 1000 rows

55 (50827,0,0,'Mobile Phone',3,6,'E wallet','Male',2,3,'Fashion',4,'Divorced',2,0,13,1,1,6,173),

56 (50828,0,NULl,'Phone',3,12,'E wallet','Male',2,3,'Laptop & Accessory',3,'Divorced',3,1,20,0,5,5,123),

57 (50829,0,10,'Phone',1,NULl,'Debit Card','Male',2,3,'Laptop & Accessory',4,'Married',0,0,10,1,1,15,123),

58 (50830,0,5,'Computer',3,14,'E wallet','Female',2,3,'Fashion',2,'Single',2,0,14,2,3,7,190),

59 (50831,0,2,'Computer',1,6,'CC','Male',2,3,'Laptop & Accessory',3,'Divorced',2,0,13,0,1,9,143),

60 (50832,0,0,'Mobile Phone',1,13,'Credit Card','Male',2,4,'Laptop & Accessory',3,'Divorced',5,0,17,1,1,0,161),

61 (50833,0,10,'Phone',1,15,'CC','Female',3,4,'Mobile',4,'Divorced',5,1,20,1,1,0,133),

62 (50834,0,13,'Mobile Phone',3,10,'E wallet','Male',3,4,'Fashion',2,'Divorced',0,0,11,2,2,11,209),

63 (50835,0,NULl,'Computer',3,0,'E wallet','Female',3,0,'Mobile',3,'Divorced',7,0,10,1,1,2,122),

64 (50836,0,0,'Phone',1,NULl,'CC','Male',3,3,'Mobile',2,'Divorced',2,0,24,0,1,2,123),

65 (50837,0,NULl,'Computer',1,15,'CC','Male',2,3,'Mobile',4,'Married',7,0,10,0,1,2,120),

66 (50838,0,3,'Phone',1,27,'CC','Male',NULl,3,'Mobile',2,'Divorced',2,0,13,0,1,2,123),

67 (50839,0,10,'Mobile Phone',1,10,'Debit Card','Male',2,2,'Fashion',3,'Married',4,1,10,1,1,4,194),

68 (50840,0,0,'Computer',1,NULl,'CC','Male',3,4,'Mobile',3,'Married',2,1,15,0,1,1,103),

69 (50841,0,13,'Mobile Phone',1,17,'Debit Card','Female',2,3,'Others',4,'Single',4,0,NULl,2,4,4,299),

70 (50842,0,14,'Phone',1,NULl,'Debit Card','Male',2,3,'Mobile',3,'Married',5,0,12,1,1,0,125),

71 (50843,0,3,'Mobile Phone',3,11,'Debit Card','Female',2,3,'Fashion',2,'Divorced',3,0,12,2,7,4,194),

72 (50844,0,11,'Mobile Phone',3,6,'Debit Card','Female',2,4,'Laptop & Accessory',2,'Single',5,0,22,1,1,1,107),

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Action Output

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrder
50065	0	24	Mobile Phone	1	9	Credit Card	Female	3	3	Grocery
50066	0	25	Mobile Phone	1	6	Debit Card	Male	3	4	Others
50067	0	15	Phone	1	15	CC	Female	2	3	Mobile
50068	1	0	Mobile Phone	1	29	Debit Card	Male	2	4	Laptop & Access

customer_churn 16 x

Action Output

#	Time	Action	Message	Duration / Fetch
31	12:51:58	UPDATE customer_churn SET HourSpendOnApp = (SELECT ROUND(AVG(HourSpendOnApp)) FROM (SEL...	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0	0.000 sec
32	12:52:04	SELECT COUNT(*) AS Total_Rows, COUNT(WarehouseToHome) AS WarehouseToHome_NotNull, COUNT...	1 row(s) returned	0.000 sec / 0.000 sec
33	12:53:24	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.015 sec

12:56 PM 15-09-2026

Impute mean OrderAmountHikeFromlastYear

The screenshot shows the MySQL Workbench interface with a SQL script in the 'SQL File 1*' editor. The script is titled 'E-Commerce Customer churn db' and contains the following queries:

```
5691 (SELECT ROUND(AVG(hourSpendOnApp))
5692 FROM (SELECT hourSpendOnApp
5693 FROM customer_churn
5694 WHERE hourSpendOnApp IS NOT NULL) AS temp);
5695
5696
5697 UPDATE customer_churn
5698 SET hourSpendOnApp = (SELECT ROUND(AVG(hourSpendOnApp))
5699 FROM (SELECT hourSpendOnApp
5700 FROM customer_churn
5701 WHERE hourSpendOnApp IS NOT NULL) AS temp)
5702 WHERE hourSpendOnApp IS NULL;
5703
5704 (SELECT ROUND(AVG(OrderAmountHikeFromlastYear))
5705 FROM (SELECT OrderAmountHikeFromlastYear FROM customer_churn
5706 WHERE OrderAmountHikeFromlastYear IS NOT NULL) AS temp);
5707
5708 UPDATE customer_churn
```

The 'Result Grid' shows the output of the first query, which is a single row with the value 16. The 'Output' pane shows the execution progress of the queries.

The screenshot shows the MySQL Workbench interface with a SQL script in the 'SQL File 1*' editor. The script is titled 'E-Commerce Customer churn db' and contains the following queries:

```
5699 FROM (SELECT hourSpendOnApp
5700 FROM customer_churn
5701 WHERE hourSpendOnApp IS NOT NULL) AS temp)
5702 WHERE hourSpendOnApp IS NULL;
5703
5704 (SELECT ROUND(AVG(OrderAmountHikeFromlastYear))
5705 FROM (SELECT OrderAmountHikeFromlastYear FROM customer_churn
5706 WHERE OrderAmountHikeFromlastYear IS NOT NULL) AS temp);
5707
5708 UPDATE customer_churn
5709 SET OrderAmountHikeFromlastYear = (SELECT ROUND(AVG(OrderAmountHikeFromlastYear))
5710 FROM (SELECT OrderAmountHikeFromlastYear FROM customer_churn
5711 WHERE OrderAmountHikeFromlastYear IS NOT NULL) AS temp)
5712 WHERE OrderAmountHikeFromlastYear IS NULL;
5713
5714
5715
5716
```

The 'Result Grid' shows the output of the first query, which is a single row with the value 16. The 'Output' pane shows the execution progress of the queries. An error message is displayed in the 'Output' pane:

```
36 13:23:49 UPDATE customer_churn SET OrderAmountHikeFromlastYear = (SELECT ROUND(AVG(OrderAmountHikeFromlastYear))
37 13:24:08 UPDATE customer_churn SET OrderAmountHikeFromlastYear = (SELECT ROUND(AVG(OrderAmountHikeFromlastYear))
38 13:24:22 SELECT COUNT(*) AS Total_Rows, COUNT(WarehouseToHome) AS WarehouseToHome_Count, COUNT(WarehouseToHome_IsNull) AS WarehouseToHome_IsNull_Count;
```

The error message is: "Error Code: 1054 Unknown column 'OrderAmountHikeFromlastYear' in 'field list'". The message indicates that the column 'OrderAmountHikeFromlastYear' does not exist in the table.

Impute mean DaySinceLastOrder

MySQL Workbench interface showing a SQL query in the SQL File 1* editor. The query is for the 'ecommerce' database, specifically the 'customer_churn' table. The query calculates the average 'DaySinceLastOrder' for customers who have not placed an order in the last year and imputes this value into the 'DaySinceLastOrder' column.

```
5762 WHERE hoursSpentOnApp IS NULL;
5763
5764 • (SELECT ROUND(AVG(OrderAmountBikeFromLastYear))
5765 FROM (SELECT OrderAmountBikeFromLastYear FROM customer_churn
5766 WHERE OrderAmountBikeFromLastYear IS NOT NULL) AS temp);
5767
5768 • UPDATE customer_churn
5769 SET OrderAmountBikeFromLastYear = (SELECT ROUND(AVG(OrderAmountBikeFromLastYear))
5770 FROM (SELECT OrderAmountBikeFromLastYear FROM customer_churn
5771 WHERE OrderAmountBikeFromLastYear IS NOT NULL) AS temp)
5772 WHERE OrderAmountBikeFromLastYear IS NULL;
5773
5774 • (SELECT ROUND(AVG(DaySinceLastOrder))
5775 FROM (SELECT DaySinceLastOrder FROM customer_churn
5776 WHERE DaySinceLastOrder IS NOT NULL) AS temp);
5777
5778
5779
```

The Result Grid shows the output of the query, which is a single row with the value 5.

Result Grid
ROUND(AVG(DaySinceLastOrder))
5

The Action Output pane shows the execution progress and messages. The final message indicates that the query was successful and returned 1 row(s).

#	Time	Action	Message	Duration / Fetch
38	13:24:22	SELECT COUNT(*) AS Total_Rows, COUNT(WarehouseToHome) AS WarehouseToHome_NotNull, COUNT...	1 row(s) returned	0.016 sec / 0.000 sec
39	13:28:08	(SELECT ROUND(AVG(DaySinceLastOrder)) FROM (SELECT DaySinceLastOrder FROM ecommerce_custom...	Error Code: 1146 Table 'ecommerce_customer_churn' doesn't exist	0.031 sec
40	13:28:19	(SELECT ROUND(AVG(DaySinceLastOrder)) FROM (SELECT DaySinceLastOrder FROM customer_churn W...	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench interface showing a SQL query in the SQL File 1* editor. The query is for the 'ecommerce' database, specifically the 'customer_churn' table. The query calculates the average 'DaySinceLastOrder' for customers who have not placed an order in the last year and imputes this value into the 'DaySinceLastOrder' column.

```
5773
5774 • (SELECT ROUND(AVG(DaySinceLastOrder))
5775 FROM (SELECT DaySinceLastOrder FROM customer_churn
5776 WHERE DaySinceLastOrder IS NOT NULL) AS temp);
5777
5778
5779 • UPDATE customer_churn
5780 SET DaySinceLastOrder = (SELECT ROUND(AVG(DaySinceLastOrder))
5781 FROM (SELECT DaySinceLastOrder FROM customer_churn
5782 WHERE DaySinceLastOrder IS NOT NULL) AS temp)
5783 WHERE DaySinceLastOrder IS NULL;
5784
5785
5786
5787
5788
5789
```

The Result Grid shows the output of the query, which is a single row with the value 5.

Result Grid
ROUND(AVG(DaySinceLastOrder))
5

The Action Output pane shows the execution progress and messages. The final message indicates that the query was successful and returned 1 row(s).

#	Time	Action	Message	Duration / Fetch
40	13:28:19	(SELECT ROUND(AVG(DaySinceLastOrder)) FROM (SELECT DaySinceLastOrder FROM customer_churn W...	1 row(s) returned	0.000 sec / 0.000 sec
41	13:29:42	UPDATE customer_churn SET DaySinceLastOrder = (SELECT ROUND(AVG(DaySinceLastOrder)) FROM (SE...	307 row(s) affected Rows matched: 307 Changed: 307 Warnings: 0	0.016 sec
42	13:30:17	SELECT COUNT(*) AS Total_Rows, COUNT(WarehouseToHome) AS WarehouseToHome_NotNull, COUNT...	1 row(s) returned	0.015 sec / 0.000 sec

- Impute mode for the following columns: Tenure, CouponUsed, OrderCount

Tenure

MySQL Workbench interface showing the following SQL queries in the SQL File 1* editor:

```

5722 WHERE daysInLastOrder IS NOT NULL) AS temp)
5723 WHERE daysInLastOrder IS NULL;
5724
5725 #Impute mode for the following columns: Tenure,
5726
5727 SELECT Tenure,COUNT(*) AS Frequency
5728 FROM customer_churn
5729 WHERE Tenure IS NOT NULL
5730 GROUP BY Tenure
5731 ORDER BY Frequency DESC
5732 LIMIT 1;
5733
5734
5735 UPDATE customer_churn
5736 SET Tenure = 1
5737 WHERE Tenure IS NULL;
5738
5739

```

The Result Grid shows the output of the first query:

Tenure	Frequency
1	690

The Action Output shows the execution of the queries:

#	Time	Action	Message	Duration / Fetch
42	13:30:17	SELECT COUNT(*) AS Total_Rows, COUNT(WarehouseToHome) AS WarehouseToHome_NotNull, COUNT...	1 row(s) returned	0.015 sec / 0.000 sec
43	13:32:38	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.016 sec / 0.000 sec
44	14:00:39	SELECT Tenure,COUNT(*) AS Frequency FROM customer_churn WHERE Tenure IS NOT NULL GROUP BY...	1 row(s) returned	0.016 sec / 0.000 sec

MySQL Workbench interface showing the following SQL queries in the SQL File 1* editor:

```

5724
5725 #Impute mode for the following columns: Tenure,
5726
5727 SELECT Tenure,COUNT(*) AS Frequency
5728 FROM customer_churn
5729 WHERE Tenure IS NOT NULL
5730 GROUP BY Tenure
5731 ORDER BY Frequency DESC
5732 LIMIT 1;
5733
5734
5735 UPDATE customer_churn
5736 SET Tenure = 1
5737 WHERE Tenure IS NULL;
5738
5739
5740
5741

```

The Result Grid shows the output of the first query:

CustomerID	Churn	Tenure	PreferredLognDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpentOnApp	NumberOfDevicesRegistered	PreferredOrder
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Access
50002	1	1	Phone	1	8	UPI	Male	3	4	Mobile
50003	1	1	Phone	1	30	Debit Card	Male	2	4	Mobile
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Access

The Action Output shows the execution of the queries:

#	Time	Action	Message	Duration / Fetch
44	14:00:39	SELECT Tenure,COUNT(*) AS Frequency FROM customer_churn WHERE Tenure IS NOT NULL GROUP BY...	1 row(s) returned	0.016 sec / 0.000 sec
45	14:07:39	UPDATE customer_churn SET Tenure = 1 WHERE Tenure IS NULL	264 row(s) affected Rows matched: 264 Changed: 264 Warnings: 0	0.016 sec
46	14:07:55	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec

Impute mode for the following columns: CouponUsed

MySQL Workbench interface showing the initial state of the 'customer_churn' table. The 'CouponUsed' column is highlighted in the result grid, showing a value of 1 for 2105 rows. The SQL editor shows a query to select 'CouponUsed' and its frequency.

```
5733
5734
5735 UPDATE customer_churn
5736 SET Tenure = 1
5737 WHERE Tenure IS NULL;
5738
5739 #Impute mode for the following columns: CouponUsed
5740
5741 SELECT CouponUsed, COUNT(*) AS Frequency
5742 FROM customer_churn
5743 WHERE CouponUsed IS NOT NULL
5744 GROUP BY CouponUsed
5745 ORDER BY Frequency DESC
5746 LIMIT 1;
```

CouponUsed	Frequency
1	2105

Result 25 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
45	14:07:39	UPDATE customer_churn SET Tenure = 1 WHERE Tenure IS NULL	264 row(s) affected Rows matched: 264 Changed: 264 Warnings: 0	0.016 sec
46	14:07:55	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
47	14:10:56	SELECT CouponUsed, COUNT(*) AS Frequency FROM customer_churn WHERE CouponUsed IS NOT NULL	1 row(s) returned	0.015 sec / 0.000 sec

MySQL Workbench interface showing the final state of the 'customer_churn' table after imputation. The 'CouponUsed' column is highlighted in the result grid, showing a value of 1 for 256 rows. The SQL editor shows a query to select 'CouponUsed' and its frequency.

```
5737 WHERE Tenure IS NULL;
5738
5739 #Impute mode for the following columns: CouponUsed
5740
5741 SELECT CouponUsed, COUNT(*) AS Frequency
5742 FROM customer_churn
5743 WHERE CouponUsed IS NOT NULL
5744 GROUP BY CouponUsed
5745 ORDER BY Frequency DESC
5746 LIMIT 1;
5747
5748 UPDATE customer_churn
5749 SET CouponUsed = 1
5750 WHERE CouponUsed IS NULL;
5751
5752
5753
5754
```

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpentOnApp	NumberOfDevicesRegistered	PreferredOrderType
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Access
50002	1	1	Phone	1	8	UPI	Male	3	4	Mobile
50003	1	1	Phone	1	30	Debit Card	Male	2	4	Mobile
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Access

customer_churn 26 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
47	14:10:56	SELECT CouponUsed, COUNT(*) AS Frequency FROM customer_churn WHERE CouponUsed IS NOT NULL	1 row(s) returned	0.015 sec / 0.000 sec
48	16:07:07	UPDATE customer_churn SET CouponUsed = 1 WHERE CouponUsed IS NULL	256 row(s) affected Rows matched: 256 Changed: 256 Warnings: 0	0.015 sec
49	16:07:25	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec

Impute mode for the following columns: OrderCount

The screenshot shows the MySQL Workbench interface with the following details:

- Navigator:** Schemas list includes 'ecom' (selected), 'employee', 'sys', and 'testdb'.
- SQL Editor:** Contains the following SQL script:

```
5748 UPDATE customer_churn
5749 SET CouponUsed = 1
5750 WHERE CouponUsed IS NULL;
5751
5752 #Impute mode for the following columns: OrderCount
5753
5754 SELECT OrderCount, COUNT(*) AS Frequency
5755 FROM customer_churn
5756 WHERE OrderCount IS NOT NULL
5757 GROUP BY OrderCount
5758 ORDER BY Frequency DESC
5759 LIMIT 1;
```
- Result Grid:** Shows a single row with 'OrderCount' 2 and 'Frequency' 2025.
- Output:** Shows the execution results of the SQL script:

#	Time	Action	Message	Duration / Fetch
48	16:07:07	UPDATE customer_churn SET CouponUsed = 1 WHERE CouponUsed IS NULL	256 row(s) affected Rows matched: 256 Changed: 256 Warnings: 0	0.015 sec
49	16:07:25	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
50	16:11:52	SELECT OrderCount, COUNT(*) AS Frequency FROM customer_churn WHERE OrderCount IS NOT NULL GR...	1 row(s) returned	0.000 sec / 0.000 sec

The screenshot shows the MySQL Workbench interface with the following details:

- Navigator:** Schemas list includes 'ecom' (selected), 'employee', 'sys', and 'testdb'.
- SQL Editor:** Contains the following SQL script:

```
5759 WHERE CouponUsed IS NULL;
5760
5761 #Impute mode for the following columns: OrderCount
5762
5763 SELECT OrderCount, COUNT(*) AS Frequency
5764 FROM customer_churn
5765 WHERE OrderCount IS NOT NULL
5766 GROUP BY OrderCount
5767 ORDER BY Frequency DESC
5768 LIMIT 1;
5769
5770 UPDATE customer_churn
5771 SET OrderCount = 2
5772 WHERE OrderCount IS NULL;
```
- Result Grid:** Shows a table with columns: CustomerID, Churn, Tenure, PreferredLoginDevice, CityTier, WarehouseToHome, PreferredPaymentMode, Gender, HourSpendOnApp, NumberOfDevicesRegistered, PreferredOrderType. The first row is highlighted.
- Output:** Shows the execution results of the SQL script:

#	Time	Action	Message	Duration / Fetch
50	16:11:52	SELECT OrderCount, COUNT(*) AS Frequency FROM customer_churn WHERE OrderCount IS NOT NULL GR...	1 row(s) returned	0.000 sec / 0.000 sec
51	16:13:56	UPDATE customer_churn SET OrderCount = 2 WHERE OrderCount IS NULL	258 row(s) affected Rows matched: 258 Changed: 258 Warnings: 0	0.015 sec
52	16:14:24	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec

- Handle outliers in the 'WarehouseToHome' column by deleting rows where the values are greater than 100.

The screenshot shows the MySQL Workbench interface with the following SQL queries in the SQL File 1* editor:

```
5754 SELECT OrderCount, COUNT(*) AS Frequency
5755 FROM customer_churn
5756 WHERE OrderCount IS NOT NULL
5757 GROUP BY OrderCount
5758 ORDER BY Frequency DESC
5759 LIMIT 10;
5760
5761 UPDATE customer_churn
5762 SET OrderCount = 2
5763 WHERE OrderCount IS NULL;
5764
5765 #handle outliers in the 'warehouseToHome' column
5766
5767 SELECT * FROM customer_churn
5768 WHERE WarehouseToHome > 100;
5769
5770
5771
```

The Result Grid shows the following data:

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat
51330	0	25	Computer	3	126	Debit Card	Male	2	3	Mobile
54125	0	26	Computer	3	127	Debit Card	Male	3	4	Mobile Phone

The Action Output shows the following results:

#	Time	Action	Message	Duration / Fetch
51	16:13:56	UPDATE customer_churn SET OrderCount = 2 WHERE OrderCount IS NULL	258 row(s) affected Rows matched: 258 Changed: 258 Warnings: 0	0.015 sec
52	16:14:24	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
53	16:25:46	SELECT * FROM customer_churn WHERE WarehouseToHome > 100 LIMIT 0, 1000	2 row(s) returned	0.015 sec / 0.000 sec

The screenshot shows the MySQL Workbench interface with the following SQL queries in the SQL File 1* editor:

```
5758 ORDER BY Frequency DESC
5759 LIMIT 10;
5760
5761 UPDATE customer_churn
5762 SET OrderCount = 2
5763 WHERE OrderCount IS NULL;
5764
5765 #handle outliers in the 'warehouseToHome' column
5766
5767 SELECT * FROM customer_churn
5768 WHERE WarehouseToHome > 100;
5769
5770 DELETE FROM customer_churn
5771 WHERE WarehouseToHome > 100;
5772
5773
5774
5775
5776
5777
5778
5779
5780
5781
5782
5783
```

The Action Output shows the following results:

#	Time	Action	Message	Duration / Fetch
52	16:14:24	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
53	16:25:46	SELECT * FROM customer_churn WHERE WarehouseToHome > 100 LIMIT 0, 1000	2 row(s) returned	0.015 sec / 0.000 sec
54	16:27:46	DELETE FROM customer_churn WHERE WarehouseToHome > 100	2 row(s) affected	0.016 sec

Dealing with Inconsistencies:

- Replace occurrences of “Phone” in the 'PreferredLoginDevice' column and “Mobile” in the 'PreferedOrderCat' column with “Mobile Phone” to ensure uniformity.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
5771 WHERE warehouseToName > 100;  
5772  
5773 SELECT * FROM customer_churn;  
5774  
5775 UPDATE customer_churn  
5776 SET PreferredLoginDevice = 'Mobile Phone'  
5777 WHERE PreferredLoginDevice = 'Phone';  
5778  
5779  
5780  
5781  
5782
```

The Result Grid shows the following data:

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToName	PreferredPaymentMode	Gender	HourSpentOnApp	NumberOfDeviceRegistered	PreferredOrderCat
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Access
50002	1	1	Phone	1	8	UPI	Male	3	4	Mobile
50003	1	1	Phone	1	30	Debit Card	Male	2	4	Mobile
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Access
50005	1	0	Phone	1	12	CC	Male	3	3	Mobile
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone
50007	1	1	Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Access
50008	1	1	Phone	1	6	CC	Male	3	3	Mobile
50009	1	13	Phone	3	9	E wallet	Male	3	4	Mobile
50010	1	1	Phone	1	31	Debit Card	Male	2	5	Mobile
50011	1	4	Mobile Phone	1	18	Debit Card	Female	3	1	Mobile

The Output pane shows the following messages:

```
55 18:26:48 SELECT * FROM customer_churn LIMIT 0, 1000 1000 row(s) returned 0.016 sec / 0.000 sec  
56 18:28:09 UPDATE customer_churn SET PreferredLoginDevice = 'Mobile Phone' WHERE PreferredLoginDevice = 'Phone' Error Code: 1146, Table 'ecommerce.ecommerce_customer_churn' doesn't exist 0.015 sec  
57 18:28:57 SELECT * FROM customer_churn LIMIT 0, 1000 1000 row(s) returned 0.000 sec / 0.000 sec
```

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
5767 SELECT * FROM customer_churn  
5768 WHERE warehouseToName > 100;  
5769  
5770 DELETE FROM customer_churn  
5771 WHERE warehouseToName > 100;  
5772  
5773 SELECT * FROM customer_churn;  
5774  
5775 UPDATE customer_churn  
5776 SET PreferredLoginDevice = 'Mobile Phone'  
5777 WHERE PreferredLoginDevice = 'Phone';  
5778
```

The Result Grid shows the following data:

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToName	PreferredPaymentMode	Gender	HourSpentOnApp	NumberOfDeviceRegistered	PreferredOrderCat
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Access
50002	1	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile
50003	1	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile
50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Access
50005	1	0	Mobile Phone	1	12	CC	Male	3	3	Mobile
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone
50007	1	1	Mobile Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Access
50008	1	1	Mobile Phone	1	6	CC	Male	3	3	Mobile
50009	1	13	Mobile Phone	3	9	E wallet	Male	3	4	Mobile
50010	1	1	Mobile Phone	1	31	Debit Card	Male	2	5	Mobile
50011	1	4	Mobile Phone	1	18	Debit Card	Female	3	1	Mobile

The Output pane shows the following messages:

```
57 18:28:57 SELECT * FROM customer_churn LIMIT 0, 1000 1000 row(s) returned 0.000 sec / 0.000 sec  
58 18:30:00 UPDATE customer_churn SET PreferredLoginDevice = 'Mobile Phone' WHERE PreferredLoginDevice = 'Phone' 1231 row(s) affected Rows matched: 1231 Changed: 1231 Warnings: 0 0.047 sec  
59 18:30:03 SELECT * FROM customer_churn LIMIT 0, 1000 1000 row(s) returned 0.016 sec / 0.000 sec
```

- Standardize payment mode values: Replace "COD" with "Cash on Delivery" and "CC" with "Credit Card" in the PreferredPaymentMode column.

COD→Cash on Delivery

The screenshot shows the MySQL Workbench interface with the following SQL queries in the SQL File 1* editor:

```

5773 SELECT * FROM customer_churn;
5774
5775 UPDATE customer_churn
5776 SET PreferredLoginDevice = 'Mobile Phone'
5777 WHERE PreferredLoginDevice = 'Phone';
5778
5779 #standardize payment mode values
5780
5781 UPDATE customer_churn
5782 SET PreferredPaymentMode = 'Cash on Delivery'
5783 WHERE PreferredPaymentMode = 'COD';
5784

```

The Result Grid shows the following data:

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDevicesRegistered	PreferredOrder
50010	1	1	Mobile Phone	1	31	Debit Card	Male	2	5	Mobile
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion
50013	1	0	Mobile Phone	1	11	COD	Male	2	3	Mobile
50014	1	0	Mobile Phone	1	15	CC	Male	3	4	Mobile
50015	1	9	Mobile Phone	3	15	Credit Card	Male	3	4	Fashion
50016	1	1	Mobile Phone	2	12	UPI	Male	3	3	Mobile
50017	1	0	Computer	1	12	Debit Card	Female	3	4	Mobile
50018	1	0	Mobile Phone	3	11	E wallet	Male	2	4	Laptop & Access
50019	1	0	Computer	1	13	Debit Card	Male	3	5	Laptop & Access
50020	1	10	Mobile Phone	1	30	Debit Card	Female	2	4	Mobile Phone

The Output pane shows the following messages:

```

57 18:28:57 SELECT * FROM customer_churn LIMIT 0, 1000
1000 row(s) returned
Duration / Fetch: 0.000 sec / 0.000 sec

58 18:30:00 UPDATE customer_churn SET PreferredLoginDevice = 'Mobile Phone' WHERE PreferredLoginDevice = 'Phone'
1231 row(s) affected Rows matched: 1231 Changed: 1231 Warnings: 0
Duration / Fetch: 0.047 sec / 0.000 sec

59 18:30:03 SELECT * FROM customer_churn LIMIT 0, 1000
1000 row(s) returned
Duration / Fetch: 0.016 sec / 0.000 sec

```

The screenshot shows the MySQL Workbench interface with the following SQL queries in the SQL File 1* editor:

```

5779 #standardize payment mode values
5780
5781 UPDATE customer_churn
5782 SET PreferredPaymentMode = 'Cash on Delivery'
5783 WHERE PreferredPaymentMode = 'COD';
5784
5785
5786
5787
5788
5789
5790

```

The Result Grid shows the following data:

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDevicesRegistered	PreferredOrder
50010	1	1	Mobile Phone	1	31	Debit Card	Male	2	5	Mobile
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion
50013	1	0	Mobile Phone	1	11	Cash on Delivery	Male	2	3	Mobile
50014	1	0	Mobile Phone	1	15	CC	Male	3	4	Mobile
50015	1	9	Mobile Phone	3	15	Credit Card	Male	3	4	Fashion
50016	1	1	Mobile Phone	2	12	UPI	Male	3	3	Mobile
50017	1	0	Computer	1	12	Debit Card	Female	3	4	Mobile
50018	1	0	Mobile Phone	3	11	E wallet	Male	2	4	Laptop & Access
50019	1	0	Computer	1	13	Debit Card	Male	3	5	Laptop & Access
50020	1	10	Mobile Phone	1	30	Debit Card	Female	2	4	Mobile Phone

The Output pane shows the following messages:

```

59 18:30:03 SELECT * FROM customer_churn LIMIT 0, 1000
1000 row(s) returned
Duration / Fetch: 0.016 sec / 0.000 sec

60 18:36:03 UPDATE customer_churn SET PreferredPaymentMode = 'Cash on Delivery' WHERE PreferredPaymentMode = 'COD'
365 row(s) affected Rows matched: 365 Changed: 365 Warnings: 0
Duration / Fetch: 0.016 sec / 0.000 sec

61 18:36:14 SELECT * FROM customer_churn LIMIT 0, 1000
1000 row(s) returned
Duration / Fetch: 0.000 sec / 0.000 sec

```

CC → Credit Card

MySQL Workbench

Local instance MySQL80 x

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Information

Schema: ecomm

SQL File 1* E-Commerce Customer churn db E-Commerce Customer churn d...

Limit to 1000 rows

5779 #standardize payment mode values

5780

5781 • UPDATE customer_churn

5782 SET PreferredPaymentMode = 'Cash on Delivery'

5783 WHERE PreferredPaymentMode = 'CCO';

5784

5785 • UPDATE customer_churn

5786 SET PreferredPaymentMode = 'Credit Card'

5787 WHERE PreferredPaymentMode = 'CC';

5788

5789

5790

Result Grid

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDevicesRegistered	PreferredOrder
50010	1	1	Mobile Phone	1	31	Debit Card	Male	2	5	Mobile
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion
50013	1	0	Mobile Phone	1	11	Cash on Delivery	Male	2	3	Mobile
50014	1	0	Mobile Phone	1	15	CC	Male	3	4	Mobile
50015	1	9	Mobile Phone	3	15	Credit Card	Male	3	4	Fashion
50016	1	1	Mobile Phone	2	12	LPT	Male	3	3	Mobile
50017	1	0	Computer	1	12	Debit Card	Female	3	4	Mobile
50018	1	0	Mobile Phone	3	11	E wallet	Male	2	4	Laptop & Access
50019	1	0	Computer	1	13	Debit Card	Male	3	5	Laptop & Access
50020	1	0	Mobile Phone	1	11	Debit Card	Female	1	1	Mobile Phone

customer_churn 33 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
59	18:30:03	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.016 sec / 0.000 sec
60	18:36:03	UPDATE customer_churn SET PreferredPaymentMode = 'Cash on Delivery' WHERE PreferredPaymentMode = 'CCO'	365 row(s) affected Rows matched: 365 Changed: 365 Warnings: 0	0.016 sec
61	18:36:14	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

25°

Search

ENG IN

06:39 PM 15-09-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

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Schema: ecomm

SQL File 1* E-Commerce Customer churn db E-Commerce Customer churn d...

Limit to 1000 rows

5768 WHERE WarehouseToHome > 100;

5769

5770 • DELETE FROM customer_churn

5771 WHERE WarehouseToHome > 100;

5772

5773 • SELECT * FROM customer_churn;

5774

5775 • UPDATE customer_churn

5776 SET PreferredLoginDevice = 'Mobile Phone'

5777 WHERE PreferredLoginDevice = 'Phone';

5778

5779 #standardize payment mode values

Result Grid

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDevicesRegistered	PreferredOrder
50009	1	13	Mobile Phone	3	9	E wallet	Male	3	4	Mobile
50010	1	1	Mobile Phone	1	31	Debit Card	Male	2	5	Mobile
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion
50013	1	0	Mobile Phone	1	11	Cash on Delivery	Male	2	3	Mobile
50014	1	0	Mobile Phone	1	15	Credit Card	Male	3	4	Mobile
50015	1	9	Mobile Phone	3	15	Credit Card	Male	3	4	Fashion
50016	1	1	Mobile Phone	2	12	LPT	Male	3	3	Mobile
50017	1	0	Computer	1	12	Debit Card	Female	3	4	Mobile
50018	1	0	Mobile Phone	3	11	E wallet	Male	2	4	Laptop & Access
50019	1	0	Computer	1	13	Debit Card	Male	3	5	Laptop & Access

customer_churn 34 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
61	18:36:14	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
62	18:40:10	UPDATE customer_churn SET PreferredPaymentMode = 'Credit Card' WHERE PreferredPaymentMode = 'CC'	273 row(s) affected Rows matched: 273 Changed: 273 Warnings: 0	0.016 sec
63	18:40:17	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

25°

Search

ENG IN

06:40 PM 15-09-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Data Transformation:

Column Renaming: -

- Rename the column "PreferedOrderCat" to "PreferredOrderCat".

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following queries:

```
SET PreferredPaymentMode = 'Cash on Delivery'
WHERE PreferredPaymentMode = 'COD';

UPDATE customer_churn
SET PreferredPaymentMode = 'Credit Card'
WHERE PreferredPaymentMode = 'CC';

--column renaming
ALTER TABLE customer_churn
RENAME COLUMN PreferedOrderCat TO PreferredOrderCat;
```

The Result Grid displays data from the `customer_churn` table:

Mode	Gender	HourSpentOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountInKilofromlastYear	Churn
Female	3	3	Laptop & Accessory	2	Single	9	1	11	1	1
Male	3	4	Mobile	3	Single	7	1	15	0	0
Male	2	4	Mobile	3	Single	6	1	14	0	0
Male	2	4	Laptop & Accessory	5	Single	8	0	23	0	0

The Action Output shows the execution of the queries:

- 66 10:55:19 SELECT * FROM customer_churn LIMIT 0, 10000. 5628 row(s) returned. Duration / Fetch: 0.000 sec / 0.000 sec
- 67 10:55:59 SELECT COUNT(*) AS Total_Rows, COUNT(WarehouseToHome) AS WarehouseToHome_NotNull, COUNT(WarehouseToHome) AS WarehouseToHome_Null FROM customer_churn. 1 row(s) returned. Duration / Fetch: 0.000 sec / 0.000 sec
- 68 19:14:56 SELECT * FROM customer_churn LIMIT 0, 10000. 5628 row(s) returned. Duration / Fetch: 0.015 sec / 0.000 sec

The screenshot shows the MySQL Workbench interface after the column rename query has been executed. The SQL Editor contains the following queries:

```
--column renaming
ALTER TABLE customer_churn
RENAME COLUMN PreferedOrderCat TO PreferredOrderCat;
```

The Result Grid displays data from the `customer_churn` table:

CustomerID	Churn	Tenure	PreferredOrderDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpentOnApp	NumberOfDeviceRegistered	PreferredOrder
50002	1	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile
50016	1	1	Mobile Phone	2	12	UPI	Male	3	3	Mobile
50098	0	10	Computer	1	17	UPI	Male	2	3	Fashion
50120	0	18	Mobile Phone	1	6	UPI	Male	3	2	Others

The Action Output shows the execution of the queries:

- 68 19:14:56 SELECT * FROM customer_churn LIMIT 0, 10000. 5628 row(s) returned. Duration / Fetch: 0.015 sec / 0.000 sec
- 69 19:16:32 ALTER TABLE customer_churn RENAME COLUMN PreferedOrderCat TO PreferredOrderCat. 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0. Duration / Fetch: 0.093 sec
- 70 19:16:44 SELECT * FROM customer_churn LIMIT 0, 10000. 5628 row(s) returned. Duration / Fetch: 0.000 sec / 0.031 sec

- Rename the column "HourSpendOnApp" to "HoursSpentOnApp".

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
5788 #column renaming
5789
5790
5791 ALTER TABLE customer_churn
5792 RENAME COLUMN PreferredOrderCat TO PreferredOrderCat;
5793
5794
5795
5796
5797
5798
5799
5800
5801
5802
5803
5804
5805
```

The Result Grid displays data from the 'customer_churn' table:

WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	Or
LPT	Male	3	4	Mobile	3	Single	7	1	15	
LPT	Male	3	3	Mobile	5	Married	5	1	22	
LPT	Male	2	3	Fashion	2	Married	2	1	14	
LPT	Male	3	2	Others	2	Divorced	1	0	16	

The Output pane shows the execution of the first query:

```
68 19:14:56 SELECT * FROM customer_churn LIMIT 0, 10000
69 19:16:32 ALTER TABLE customer_churn RENAME COLUMN PreferredOrderCat TO PreferredOrderCat
70 19:16:44 SELECT * FROM customer_churn LIMIT 0, 10000
```

The screenshot shows the MySQL Workbench interface after the first query. The SQL editor contains the following queries:

```
5788 #column renaming
5789
5790
5791 ALTER TABLE customer_churn
5792 RENAME COLUMN PreferredOrderCat TO PreferredOrderCat;
5793
5794 ALTER TABLE customer_churn
5795 RENAME COLUMN HourSpendOnApp TO HoursSpentOnApp;
5796
5797
5798
5799
5800
5801
5802
5803
5804
5805
```

The Result Grid displays data from the 'customer_churn' table:

WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Com
6	Debit Card	Female	3	3	Laptop & Accessory	2	Single	9	1
8	LPT	Male	3	4	Mobile	3	Single	7	1
30	Debit Card	Male	2	4	Mobile	3	Single	6	1
15	Debit Card	Male	2	4	Laptop & Accessory	5	Single	8	0

The Output pane shows the execution of the second query:

```
70 19:16:44 SELECT * FROM customer_churn LIMIT 0, 10000
71 19:20:25 ALTER TABLE customer_churn RENAME COLUMN HourSpendOnApp TO HoursSpentOnApp
72 19:20:31 SELECT * FROM customer_churn LIMIT 0, 10000
```

Creating New Columns:

- Create a new column named 'ComplaintReceived' with values "Yes" if the corresponding value in the 'Complain' is 1, and "No" otherwise.

The screenshot shows the MySQL Workbench interface with the 'E-Commerce Customer churn db' selected. The 'customer_churn' table is displayed in the 'Result Grid' with the following columns: SatisfactionScore, MaritalStatus, NumberOfAddress, Complain, OrderAmountSinceFromLastYear, CouponUsed, OrderCount, DaySinceLastOrder, CashbackAmount, and ComplaintReceived. The 'ComplaintReceived' column is currently empty. The 'Action Output' pane shows the execution of a query that added the 'ComplaintReceived' column as a VARCHAR(3) type.

SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountSinceFromLastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplaintReceived
9	Single	1	11	1	1	5	160	160	
7	Single	1	15	0	1	0	121	121	
6	Single	1	14	0	1	3	120	120	
8	Single	0	23	0	1	3	134	134	

The screenshot shows the MySQL Workbench interface after the 'ComplaintReceived' column has been added and updated. The 'Result Grid' now shows the 'ComplaintReceived' column with values 'Yes' or 'No' based on the 'Complain' column. The 'Action Output' pane shows the execution of a query that updated the 'ComplaintReceived' column using a CASE statement.

SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountSinceFromLastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplaintReceived
9	Single	1	11	1	1	5	160	160	Yes
7	Single	1	15	0	1	0	121	121	Yes
6	Single	1	14	0	1	3	120	120	Yes
8	Single	0	23	0	1	3	134	134	No
3	Single	0	11	1	1	3	130	130	No
2	Single	1	22	4	6	7	139	139	Yes
4	Divorced	0	14	0	1	0	121	121	No
3	Divorced	1	16	2	2	0	123	123	Yes
2	Divorced	2	1	0	1	2	127	127	Yes
2	Single	0	12	1	1	1	123	123	No
2	Divorced	2	0	16	9	15	295	295	No
10	Single	1	13	0	1	0	154	154	Yes
2	Single	1	13	2	2	2	134	134	Yes

- Create a new column named 'ChurnStatus'. Set its value to “Churned” if the corresponding value in the 'Churn' column is 1, else assign “Active”

The screenshot shows the MySQL Workbench interface with the following SQL queries in the SQL File 1* editor:

```
5799 ALTER TABLE customer_churn
5800 ADD COLUMN ComplainedReceived VARCHAR(3);
5801
5802 UPDATE customer_churn
5803 SET ComplainedReceived = CASE
5804 WHEN Churn = 1 THEN 'Yes'
5805 ELSE 'No'
5806 END;
5807
5808 ALTER TABLE customer_churn
5809 ADD COLUMN ChurnStatus VARCHAR(10);
5810
5811
5812
5813
5814
5815
```

The Result Grid shows the following data:

MaritalStatus	NumberOfAddress	Complain	OrderAmountSinceFromLastYear	CouponUsed	OrderCount	DaysSinceLastOrder	CashbackAmount	ComplainReceived	ChurnStatus
Single	9	1	11	1	1	5	160	Yes	Churned
Single	7	1	15	0	1	0	121	Yes	Churned
Single	6	1	14	0	1	3	120	Yes	Churned
Single	8	0	23	0	1	3	134	No	Churned
Divorced	3	0	11	1	1	3	130	No	Churned

The Action Output shows the following messages:

```
76 19:35:04 SELECT * FROM customer_churn LIMIT 0, 10000 5628 row(s) returned 0.015 sec / 0.000 sec
77 19:40:45 ALTER TABLE customer_churn ADD COLUMN ChurnStatus VARCHAR(10) 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.047 sec
78 19:40:54 SELECT * FROM customer_churn LIMIT 0, 10000 5628 row(s) returned 0.032 sec / 0.000 sec
```

The screenshot shows the MySQL Workbench interface with the following SQL queries in the SQL File 1* editor:

```
5809 ADD COLUMN ChurnStatus VARCHAR(10);
5810
5811 UPDATE customer_churn
5812 SET ChurnStatus = CASE
5813 WHEN Churn = 1 THEN 'Churned'
5814 ELSE 'Active'
5815 END;
5816
5817
```

The Result Grid shows the following data:

MaritalStatus	NumberOfAddress	Complain	OrderAmountSinceFromLastYear	CouponUsed	OrderCount	DaysSinceLastOrder	CashbackAmount	ComplainReceived	ChurnStatus
Single	9	1	11	1	1	5	160	Yes	Churned
Single	7	1	15	0	1	0	121	Yes	Churned
Single	6	1	14	0	1	3	120	Yes	Churned
Single	8	0	23	0	1	3	134	No	Churned
Single	3	0	11	1	1	3	130	No	Churned
Single	2	1	22	4	6	7	139	Yes	Churned
Divorced	4	0	14	0	1	0	121	No	Churned
Divorced	3	1	16	2	2	0	123	Yes	Churned
Divorced	2	1	14	0	1	2	127	Yes	Churned
Single	2	0	12	1	1	1	123	No	Churned
Divorced	2	0	16	9	15	8	295	No	Churned
Single	10	1	13	0	1	0	134	Yes	Churned
Single	2	1	13	2	2	2	134	Yes	Churned
Divorced	1	1	17	0	1	0	134	Yes	Churned

The Action Output shows the following messages:

```
78 19:40:54 SELECT * FROM customer_churn LIMIT 0, 10000 5628 row(s) returned 0.032 sec / 0.000 sec
79 19:43:49 UPDATE customer_churn SET ChurnStatus = CASE WHEN Churn = 1 THEN 'Churned' ELSE 'Active' END 5628 row(s) affected Rows matched: 5628 Changed: 5628 Warnings: 0 0.234 sec
80 19:43:57 SELECT * FROM customer_churn LIMIT 0, 10000 5628 row(s) returned 0.000 sec / 0.016 sec
```

Column Dropping:

- Drop the columns "Churn" and "Complain" from the table.

MySQL Workbench interface showing the SQL Editor with the following queries:

```
5800 ADD COLUMN ComplainReceived VARCHAR(3);
5801
5802 UPDATE customer_churn
5803 SET ComplainReceived = CASE
5804 WHEN Churn = 1 THEN 'yes'
5805 ELSE 'no'
5806 END;
5807
5808 ALTER TABLE customer_churn
5809 ADD COLUMN ChurnStatus VARCHAR(10);
5810
5811 UPDATE customer_churn
5812 SET ChurnStatus = CASE
5813 WHEN Churn = 1 THEN 'churned'
5814 ELSE 'Active'
5815 END;
5816
5817 #DROP THE COLUMN NAMED "Churn" and "Complain"
```

The Result Grid shows the following data:

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDevicesRegistered	PreferredOrderCat
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Acces
50002	1	1	Mobile Phone	1	8	LPI	Male	3	4	Mobile
50003	1	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile
50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Acces

The Output pane shows the following actions:

- 78 19:40:54 SELECT * FROM customer_churn LIMIT 0, 10000. Message: 5628 row(s) returned. Duration / Fetch: 0.032 sec / 0.000 sec.
- 79 19:43:49 UPDATE customer_churn SET ChurnStatus = CASE WHEN Churn = 1 THEN 'churned' ELSE 'Active' END. Message: 5628 row(s) affected Rows matched: 5628 Changed: 5628 Warnings: 0. Duration / Fetch: 0.234 sec / 0.000 sec.
- 80 19:43:57 SELECT * FROM customer_churn LIMIT 0, 10000. Message: 5628 row(s) returned. Duration / Fetch: 0.000 sec / 0.016 sec.

MySQL Workbench interface showing the SQL Editor with the following queries:

```
5812 SET ChurnStatus = CASE
5813 WHEN Churn = 1 THEN 'churned'
5814 ELSE 'Active'
5815 END;
5816
5817 #DROP THE COLUMN NAMED "Churn" and "Complain"
5818
5819 ALTER TABLE customer_churn
5820 DROP COLUMN Churn;
5821 DROP COLUMN Complain;
5822
5823
5824
5825
5826
5827
5828
5829
```

The Result Grid shows the following data:

CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDevicesRegistered	PreferredOrderCat	Si
50001	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2
50002	1	Mobile Phone	1	8	LPI	Male	3	4	Mobile	3
50003	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile	3
50004	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5

The Output pane shows the following actions:

- 80 19:43:57 SELECT * FROM customer_churn LIMIT 0, 10000. Message: 5628 row(s) returned. Duration / Fetch: 0.000 sec / 0.016 sec.
- 81 19:48:59 ALTER TABLE customer_churn DROP COLUMN Churn. Message: 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0. Duration / Fetch: 0.032 sec / 0.000 sec.
- 82 19:49:08 SELECT * FROM customer_churn LIMIT 0, 10000. Message: 5628 row(s) returned. Duration / Fetch: 0.015 sec / 0.016 sec.

Data Exploration and Analysis:

- Retrieve the count of churned and active customers from the dataset.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
5812 SET ChurnStatus = CASE
5813 WHEN Churn = 1 THEN 'Churned'
5814 ELSE 'Active'
5815 END;
5816
5817 DROP THE COLUMN NAMED "Churn" and "Complain"
5818
5819 ALTER TABLE customer_churn
5820 DROP COLUMN Churn;
5821 DROP COLUMN Complain;
5822
5823 #count of churned and active customers
5824
5825 SELECT ChurnStatus, COUNT(*) AS CustomerCount
5826 FROM customer_churn
5827 GROUP BY ChurnStatus;
```

The Result Grid shows the following data:

ChurnStatus	CustomerCount
Churned	948
Active	4680

The Output pane shows the execution log with the following messages:

- 83 19:55:27 SELECT ChurnStatus, COUNT(*) AS CustomerCount FROM customer_churn GROUP BY ChurnStatus Message: Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se... 0.000 sec / 0.000 sec
- 84 19:55:57 SELECT ChurnStatus, COUNT(*) AS CustomerCount FROM customer_churn GROUP BY ChurnStatus Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se... 0.000 sec / 0.000 sec
- 85 19:56:34 SELECT ChurnStatus, COUNT(*) AS CustomerCount FROM customer_churn GROUP BY ChurnStatus LIMIT 0... 2 row(s) returned 0.000 sec / 0.000 sec

- Display the average tenure and total cashback amount of customers who churned.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
5818 ALTER TABLE customer_churn
5819 DROP COLUMN Churn;
5820 DROP COLUMN Complain;
5821
5822 #count of churned and active customers
5823
5824 SELECT ChurnStatus, COUNT(*) AS CustomerCount
5825 FROM customer_churn
5826 GROUP BY ChurnStatus;
5827
5828 #average tenure and total cashback amount of customers
5829
5830 SELECT ROUND(AVG(Tenure), 2) AS AverageTenure,
5831 ROUND(SUM(CashbackAmount), 2) AS TotalCashbackAmount
5832 FROM customer_churn
5833 WHERE ChurnStatus = 'Churned';
5834
```

The Result Grid shows the following data:

AverageTenure	TotalCashbackAmount
3.18	152030

The Output pane shows the execution log with the following messages:

- 85 19:56:34 SELECT ChurnStatus, COUNT(*) AS CustomerCount FROM customer_churn GROUP BY ChurnStatus LIMIT 0... 2 row(s) returned 0.000 sec / 0.000 sec
- 86 19:59:27 SELECT ROUND(AVG(Tenure), 2) AS AverageTenure, ROUND(SUM(CashbackAmount), 2) AS TotalCashbac... Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se... 0.000 sec / 0.000 sec
- 87 20:05:48 SELECT ROUND(AVG(Tenure), 2) AS AverageTenure, ROUND(SUM(CashbackAmount), 2) AS TotalCashbac... 1 row(s) returned 0.000 sec / 0.000 sec

- Determine the percentage of churned customers who complained.

MySQL Workbench interface showing a SQL query in the 'E-Commerce Customer churn db' database. The query calculates the percentage of churned customers who complained using a CASE statement. The result grid shows the value 53.39.

```

SELECT ROUND(AVG(Tenure), 2) AS AverageTenure,
ROUND(SUM(CashbackAmount), 2) AS TotalCashbackAmount
FROM customer_churn
WHERE ChurnStatus = 'Churned';

-- Determine the percentage of churned customers who complained.
SELECT ROUND(SUM(CASE WHEN ComplaintReceived = 'Yes'
THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS ComplaintPercentage
FROM customer_churn
WHERE ChurnStatus = 'Churned';
  
```

Result Grid: ComplaintPercentage, 53.39

Output: Action Output

Message: Error Code: 1064 'You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '' at line 1' (Message: Error Code: 1146 Table 'ecommerce_customer_churn' doesn't exist)

- Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessory.

MySQL Workbench interface showing a SQL query in the 'E-Commerce Customer churn db' database. The query identifies the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessory using a GROUP BY clause. The result grid shows the value 3.

```

SELECT CityTier, COUNT(*) AS ChurnedCustomers
FROM customer_churn
WHERE ChurnStatus = 'Churned'
AND PreferredOrderCat = 'Laptop & Accessory'
GROUP BY CityTier
ORDER BY ChurnedCustomers DESC
LIMIT 1;
  
```

Result Grid: CityTier, ChurnedCustomers, 3, 150

Output: Action Output

Message: Error Code: 1064 'You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '' at line 1' (Message: Error Code: 1146 Table 'ecommerce_customer_churn' doesn't exist)

- Identify the most preferred payment mode among active customers.

The screenshot shows the MySQL Workbench interface with a query executed in the SQL Editor. The query is as follows:

```

SELECT CityTier, COUNT(*) AS ChurnedCustomers
FROM customer_churn
WHERE ChurnStatus = 'Churned'
AND PreferredOrderCat = 'Laptop & Accessory'
GROUP BY CityTier
ORDER BY ChurnedCustomers DESC
LIMIT 1;

#the most preferred payment mode among active customers

SELECT PreferredPaymentMode, COUNT(*) AS CustomerCount
FROM customer_churn
WHERE ChurnStatus = 'Active'
GROUP BY PreferredPaymentMode
ORDER BY CustomerCount DESC
LIMIT 1;

```

The Results Grid shows the following data:

PreferredPaymentMode	CustomerCount
Debit Card	1956

The Action Output pane shows the execution details of the queries.

- Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.

The screenshot shows the MySQL Workbench interface with a query executed in the SQL Editor. The query is as follows:

```

SELECT PreferredPaymentMode, COUNT(*) AS CustomerCount
FROM customer_churn
WHERE ChurnStatus = 'Active'
GROUP BY PreferredPaymentMode
ORDER BY CustomerCount DESC
LIMIT 1;

#Total Order Amount Hike for Single Customers who prefer Mobile Phones

SELECT SUM((OrderAmountHikeFromLastYear)) AS TotalOrderAmountHike
FROM customer_churn
WHERE MaritalStatus = 'Single'
AND PreferredOrderCat = 'Mobile Phone';

```

The Results Grid shows the following data:

TotalOrderAmountHike
7738

The Action Output pane shows the execution details of the queries.

- Find the average number of devices registered among customers who used UPI as their preferred payment mode.

The screenshot shows the MySQL Workbench interface with a query executed in the SQL editor. The query is as follows:

```
GROUP BY PreferredPaymentMode
ORDER BY CustomerCount DESC
LIMIT 1;

#Total Order Amount hike for Single Customers who Prefer Mobile Phones
SELECT SUM(OrderAmountHikeFromLastYear) AS TotalOrderAmountHike
FROM customer_churn
WHERE MaritalStatus = 'Single'
AND PreferredOrderCat = 'Mobile Phone';

#the average number of devices registered among customers who used UPI as preferred paymentmode
SELECT ROUND(AVG(NumberOfDeviceRegistered), 2) AS AverageDevices
FROM customer_churn
WHERE PreferredPaymentMode = 'UPI';
```

The result grid shows the output of the query:

AverageDevices
3.72

The Action Output pane shows the execution details of the query:

#	Time	Action	Message	Duration / Fetch
93	20.26.24	SELECT PreferredPaymentMode COUNT(*) AS CustomerCount FROM customer_churn WHERE ChurnStatus = 'Completed' AND PreferredPaymentMode = 'UPI';	1 row(s) returned	0.016 sec / 0.000 sec
94	20.31.19	SELECT SUM(OrderAmountHikeFromLastYear) AS TotalOrderAmountHike FROM customer_churn WHERE MaritalStatus = 'Single' AND PreferredOrderCat = 'Mobile Phone';	1 row(s) returned	0.031 sec / 0.000 sec
95	20.35.08	SELECT ROUND(AVG(NumberOfDeviceRegistered), 2) AS AverageDevices FROM customer_churn WHERE PreferredPaymentMode = 'UPI';	1 row(s) returned	0.015 sec / 0.000 sec

- Determine the city tier with the highest number of customers.

The screenshot shows the MySQL Workbench interface with a query executed in the SQL editor. The query is as follows:

```
SELECT CityTier, COUNT(*) AS CustomerCount
FROM customer_churn
GROUP BY CityTier
ORDER BY CustomerCount DESC
LIMIT 1;
```

The result grid shows the output of the query:

CityTier	CustomerCount
1	3666

The Action Output pane shows the execution details of the query:

#	Time	Action	Message	Duration / Fetch
94	20.31.19	SELECT SUM(OrderAmountHikeFromLastYear) AS TotalOrderAmountHike FROM customer_churn WHERE MaritalStatus = 'Single' AND PreferredOrderCat = 'Mobile Phone';	1 row(s) returned	0.031 sec / 0.000 sec
95	20.35.08	SELECT ROUND(AVG(NumberOfDeviceRegistered), 2) AS AverageDevices FROM customer_churn WHERE PreferredPaymentMode = 'UPI';	1 row(s) returned	0.015 sec / 0.000 sec
96	20.37.22	SELECT CityTier COUNT(*) AS CustomerCount FROM customer_churn GROUP BY CityTier ORDER BY CustomerCount DESC LIMIT 1;	1 row(s) returned	0.000 sec / 0.000 sec

- Identify the gender that utilized the highest number of coupons.

MySQL Workbench interface showing a query to identify the gender that utilized the highest number of coupons. The query is executed against the 'E-Commerce Customer churn db' database.

```

SELECT Gender, SUM(CouponUsed) AS TotalCouponsUsed
FROM customer_churn
GROUP BY Gender
ORDER BY TotalCouponsUsed DESC
LIMIT 1;

```

The result grid shows the following data:

Gender	TotalCouponsUsed
Male	5629

- List the number of customers and the maximum hours spent on the app in each preferred order category.

MySQL Workbench interface showing a query to list the number of customers and the maximum hours spent on the app in each preferred order category. The query is executed against the 'E-Commerce Customer churn db' database.

```

SELECT PreferredOrderCat, COUNT(*) AS CustomerCount,
MAX(HoursSpentOnApp) AS MaximumHoursSpent
FROM customer_churn
GROUP BY PreferredOrderCat;

```

The result grid shows the following data:

PreferredOrderCat	CustomerCount	MaximumHoursSpent
Laptop & Accessory	2050	5
Mobile	808	4
Mobile Phone	1270	5
Others	264	4
Fashion	826	5
Grocery	410	4

- Calculate the total order count for customers who prefer using credit cards and have the maximum satisfaction score.

MySQL Workbench interface showing a SQL query in the 'E-Commerce Customer churn db' database. The query is:

```
5895 MAX(HoursSpentOnApp) AS MaximumHoursSpent
5896 FROM customer_churn
5897 GROUP BY PreferredOrderCat;
5898
5899 #total order count for customers who prefer using credit cards with maximum satisfaction score.
5900
5901 SELECT SUM(OrderCount) AS TotalOrderCount
5902 FROM customer_churn
5903 WHERE PreferredPaymentMode = 'Credit Card'
5904 AND SatisfactionScore = (SELECT MAX(SatisfactionScore)
5905 FROM customer_churn);
5906
5907
```

The result grid shows the following data:

TotalOrderCount
1122

The bottom panel shows the execution log with the following messages:

#	Time	Action	Message	Duration / Fetch
1	11:26:25	SELECT SUM(OrderCount) AS TotalOrderCount FROM customer_churn WHERE PreferredPaymentMode = 'Credit Card'	Error Code: 1046. No database selected. Select the default DB to be used by double-clicking its name in the SCHEMAS list.	0.000 sec
2	11:27:15	USE ecom	0 row(s) affected	0.000 sec
3	11:27:26	SELECT SUM(OrderCount) AS TotalOrderCount FROM customer_churn WHERE PreferredPaymentMode = 'Credit Card' AND SatisfactionScore = (SELECT MAX(SatisfactionScore) FROM customer_churn);	1 row(s) returned	0.016 sec / 0.000 sec

- What is the average satisfaction score of customers who have complained?

MySQL Workbench interface showing a SQL query in the 'E-Commerce Customer churn db' database. The query is:

```
5908
5909 SELECT SUM(OrderCount) AS TotalOrderCount
5910 FROM customer_churn
5911 WHERE PreferredPaymentMode = 'Credit Card'
5912 AND SatisfactionScore = (SELECT MAX(SatisfactionScore)
5913 FROM customer_churn);
5914
5915 #average satisfaction score of customers who have complained
5916
5917 SELECT ROUND(AVG(SatisfactionScore), 2) AS AverageSatisfactionScore
5918 FROM customer_churn
5919 WHERE ComplaintReceived = 'Yes';
5920
5921
```

The result grid shows the following data:

AverageSatisfactionScore
3.00

The bottom panel shows the execution log with the following messages:

#	Time	Action	Message	Duration / Fetch
2	11:27:15	USE ecom	0 row(s) affected	0.000 sec
3	11:27:26	SELECT SUM(OrderCount) AS TotalOrderCount FROM customer_churn WHERE PreferredPaymentMode = 'Credit Card' AND SatisfactionScore = (SELECT MAX(SatisfactionScore) FROM customer_churn);	1 row(s) returned	0.016 sec / 0.000 sec
4	11:36:43	SELECT ROUND(AVG(SatisfactionScore), 2) AS AverageSatisfactionScore FROM customer_churn WHERE ComplaintReceived = 'Yes';	1 row(s) returned	0.000 sec / 0.000 sec

- List the preferred order category among customers who used more than 5 coupons.

MySQL Workbench

Local instance MySQL80

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SQL File 1* E-Commerce Customer churn db E-Commerce Customer churn d

Limit to 1000 rows

5908

5909 • SELECT ROUND(AVG(SatisfactionScore), 2) AS AverageSatisfactionScore

5910 FROM customer_churn

5911 WHERE ComplaintReceived = 'Yes';

5912

5913 #Preferred order category among customers who used more than 5 coupons.

5914

5915 • SELECT DISTINCT PreferredOrderCat

5916 FROM customer_churn

5917 WHERE CouponUsed > 5;

5918

5919

5920

Result Grid

PreferredOrderCat

Others

Laptop & Accessory

Fashion

Mobile

Grocery

Mobile Phone

Administration Schemas

Information

No object selected

customer_churn 3

Output

Action Output

Time Action Message Duration / Fetch

5 17:40:33 SELECT DISTINCT PreferredOrderCat FROM customer_churn WHERE CouponUsed > 5 LIMIT 0, 1000 Error Code: 1054. Unknown column 'PreferredOrderCat' in field list 0.016 sec / 0.000 sec

6 17:40:46 SELECT DISTINCT PreferredOrderCat FROM customer_churn WHERE CouponUsed > 5 LIMIT 0, 1000 Error Code: 1054. Unknown column 'PreferredOrderCat' in field list 0.000 sec / 0.000 sec

7 17:41:15 SELECT DISTINCT PreferredOrderCat FROM customer_churn WHERE CouponUsed > 5 LIMIT 0, 1000 6 row(s) returned 0.000 sec / 0.000 sec

Object Info Session

29°C Mostly cloudy

Search

ENG IN

05:42 PM 15-09-2026

- List the top 3 preferred order categories with the highest average cashback amount.

MySQL Workbench

Local instance MySQL80

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testdb

SQL File 1* E-Commerce Customer churn db E-Commerce Customer churn d

Limit to 1000 rows

5916 FROM customer_churn

5917 WHERE CouponUsed > 5;

5918

5919 #Top 3 preferred order categories with the highest average cashback amount.

5920

5921 • SELECT PreferredOrderCat,

5922 ROUND(AVG(CashbackAmount), 2) AS AverageCashback

5923 FROM customer_churn

5924 GROUP BY PreferredOrderCat

5925 ORDER BY AverageCashback DESC

5926 LIMIT 3;

5927

5928

5929

5930

5931

Result Grid

PreferredOrderCat AverageCashback

Others 304.45

Grocery 266.24

Fashion 210.40

Administration Schemas

Information

No object selected

Result 6

Output

Action Output

Time Action Message Duration / Fetch

9 18:32:21 SELECT PreferredOrderCat, ROUND(AVG(CashbackAmount), 2) AS AverageCashback FROM customer_churn... 3 row(s) returned 0.016 sec / 0.000 sec

10 18:32:23 SELECT PreferredOrderCat, ROUND(AVG(CashbackAmount), 2) AS AverageCashback FROM customer_churn... 3 row(s) returned 0.015 sec / 0.000 sec

11 18:32:25 SELECT PreferredOrderCat, ROUND(AVG(CashbackAmount), 2) AS AverageCashback FROM customer_churn... 3 row(s) returned 0.000 sec / 0.000 sec

Object Info Session

29°C Mostly cloudy

Search

ENG IN

06:32 PM 15-09-2026

- Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

5923 FROM customer_churn
5924 GROUP BY PreferredOrderCat
5925 ORDER BY AverageCashback DESC
5926 LIMIT 3;
5927
5928 #payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.
5929
5930 SELECT CustomerID,Tenure,OrderCount,PreferredPaymentMode
5931 FROM customer_churn
5932 WHERE Tenure = 10
5933 AND OrderCount > 500;
5934
5935
5936
5937
5938

```

The result grid is empty. The output pane shows the execution of the query:

#	Time	Action	Message	Duration / Fetch
13	18.44.17	SELECT CustomerID,Tenure,OrderCount,PreferredPaymentMode FROM customer_churn WHERE Tenure = 5 ...	0 row(s) returned	0.000 sec / 0.000 sec
14	18.44.39	SELECT CustomerID,Tenure,OrderCount,PreferredPaymentMode FROM customer_churn WHERE Tenure = 1 ...	9 row(s) returned	0.000 sec / 0.000 sec
15	18.44.47	SELECT CustomerID,Tenure,OrderCount,PreferredPaymentMode FROM customer_churn WHERE Tenure = 1 ...	0 row(s) returned	0.015 sec / 0.000 sec

- Categorize customers based on their distance from the warehouse to home such as 'Very Close Distance' for distances ≤ 5 km, 'Close Distance' for ≤ 10 km, 'Moderate Distance' for ≤ 15 km, and 'Far Distance' for > 15 km. Then, display the churn status breakdown for each distance category.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

5930 SELECT CustomerID,Tenure,OrderCount,PreferredPaymentMode
5931 FROM customer_churn
5932 WHERE Tenure = 10
5933 AND OrderCount > 500;
5934
5935 #Categorize customers based on their distance from the warehouse to home
5936
5937 SELECT * FROM customer_churn;
5938
5939 ALTER TABLE customer_churn
5940 ADD COLUMN DistanceCategory VARCHAR(50);
5941
5942
5943
5944
5945

```

The result grid shows the following data:

id	Status	NumberOfAddress	OrderAmount	TimeFromLastYear	CouponUsed	OrderCount	DaysSinceLastOrder	CashbackAmount	ComplainReceived	ChurnStatus	DistanceCategory
9	11	1	1	5	160	Yes	Churned				
7	15	0	1	0	121	Yes	Churned				
6	14	0	1	3	120	Yes	Churned				
8	23	0	1	3	134	No	Churned				

The output pane shows the execution of the query:

#	Time	Action	Message	Duration / Fetch
16	18.48.42	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
17	18.49.13	ALTER TABLE customer_churn ADD COLUMN DistanceCategory VARCHAR(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
18	18.49.18	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL80 x

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Information

No object selected

SQL File 1* E-Commerce Customer churn db E-Commerce Customer churn d...

Limit to 10000 rows

SQL

5937 SELECT * FROM customer_churn;

5938

5939 ALTER TABLE customer_churn

5940 ADD COLUMN DistanceCategory VARCHAR(50);

5941

5942 SET SQL_SAFE_UPDATES = 0;

5943

5944 UPDATE customer_churn

5945 SET DistanceCategory = CASE WHEN WarehouseToHome <= 5 THEN 'Very Close Distance'

5946 WHEN WarehouseToHome <= 10 THEN 'Close Distance'

5947 WHEN WarehouseToHome <= 15 THEN 'Moderate Distance' ELSE 'Far Distance'

5948 END;

Result Grid

#	Status	NumberOfAddress	OrderAmount	WheFromLastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplainReceived	ChurnStatus	DistanceCategory
1	9	11	1	1	1	5	160	Yes	Churned	Close Distance	
2	7	15	0	1	0	0	121	Yes	Churned	Close Distance	
3	6	14	0	1	3	120	Yes	Churned	Far Distance		
4	8	23	0	1	3	134	No	Churned	Moderate Distance		
5	3	11	1	1	3	130	No	Churned	Moderate Distance		
6	2	22	4	6	7	139	Yes	Churned	Far Distance		
7	ed	4	14	0	1	0	121	No	Churned	Moderate Distance	
8	ed	3	16	2	2	0	123	Yes	Churned	Close Distance	
9	ed	2	14	0	1	2	127	Yes	Churned	Close Distance	

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	22:38:47	USE ecom	0 row(s) affected	0.015 sec
4	22:38:57	UPDATE customer_churn SET DistanceCategory = CASE WHEN WarehouseToHome <= 5 THEN 'Very Close Distance' WHEN WarehouseToHome <= 10 THEN 'Close Distance' WHEN WarehouseToHome <= 15 THEN 'Moderate Distance' ELSE 'Far Distance' END;	5628 row(s) affected Rows matched: 5628 Changed: 5628 Warnings: 0	0.438 sec
5	22:39:09	SELECT * FROM customer_churn LIMIT 0, 10000	5628 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

32°C Sunny

Search

ENG IN

10:41 PM 15-09-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

Local instance MySQL80 x

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Information

No object selected

SQL File 1* E-Commerce Customer churn db E-Commerce Customer churn d...

Limit to 10000 rows

SQL

5943

5944 UPDATE customer_churn

5945 SET DistanceCategory = CASE WHEN WarehouseToHome <= 5 THEN 'Very Close Distance'

5946 WHEN WarehouseToHome <= 10 THEN 'Close Distance'

5947 WHEN WarehouseToHome <= 15 THEN 'Moderate Distance' ELSE 'Far Distance'

5948 END;

5949

5950 SELECT WarehouseToHome, DistanceCategory

5951 FROM customer_churn;

5952

5953

Result Grid

#	WarehouseToHome	DistanceCategory
1	6	Close Distance
2	8	Close Distance
3	30	Far Distance
4	15	Moderate Distance
5	12	Moderate Distance
6	22	Far Distance
7	11	Moderate Distance
8	6	Close Distance
9	9	Close Distance
10	31	Far Distance
11	18	Far Distance

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	22:38:57	UPDATE customer_churn SET DistanceCategory = CASE WHEN WarehouseToHome <= 5 THEN 'Very Close Distance' WHEN WarehouseToHome <= 10 THEN 'Close Distance' WHEN WarehouseToHome <= 15 THEN 'Moderate Distance' ELSE 'Far Distance' END;	5628 row(s) affected Rows matched: 5628 Changed: 5628 Warnings: 0	0.438 sec
5	22:39:09	SELECT * FROM customer_churn LIMIT 0, 10000	5628 row(s) returned	0.000 sec / 0.000 sec
6	22:47:49	SELECT WarehouseToHome, DistanceCategory FROM customer_churn LIMIT 0, 10000	5628 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

32°C Sunny

Search

ENG IN

10:48 PM 15-09-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

- List the customer's order details who are married, live in City Tier-1, and their order counts are more than the average number of orders placed by all customers.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

SELECT * FROM customer_churn
WHERE MaritalStatus = 'Married'
AND CityTier = 1 AND OrderCount > (SELECT AVG(OrderCount) FROM customer_churn);

```

The query results are displayed in a table with the following columns: CityTier, WarehouseToHome, PreferredPaymentMode, Gender, HoursSpentOnApp, NumberOfDeviceRegistered, PreferredOrderCat, SatisfactionScore, MaritalStatus, and NumberOfAddr. The results show 10 rows of data for married customers in City Tier 1 whose order count is greater than the average.

The output pane at the bottom shows the execution progress:

#	Time	Action	Message	Duration / Fetch
5	22:39:09	SELECT * FROM customer_churn LIMIT 0, 10000	5628 row(s) returned	0.000 sec / 0.000 sec
6	22:47:49	SELECT WarehouseToHome, DistanceCategory FROM customer_churn LIMIT 0, 10000	5628 row(s) returned	0.000 sec / 0.000 sec
7	22:55:18	SELECT * FROM customer_churn WHERE MaritalStatus = 'Married' AND CityTier = 1 AND OrderCount > (SEL...	576 row(s) returned	0.016 sec / 0.000 sec

- A) Create a 'customer_returns' table in the 'ecommm' database and insert the

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```

SELECT * FROM customer_churn
WHERE MaritalStatus = 'Married'
AND CityTier = 1 AND OrderCount > (SELECT AVG(OrderCount) FROM customer_churn);

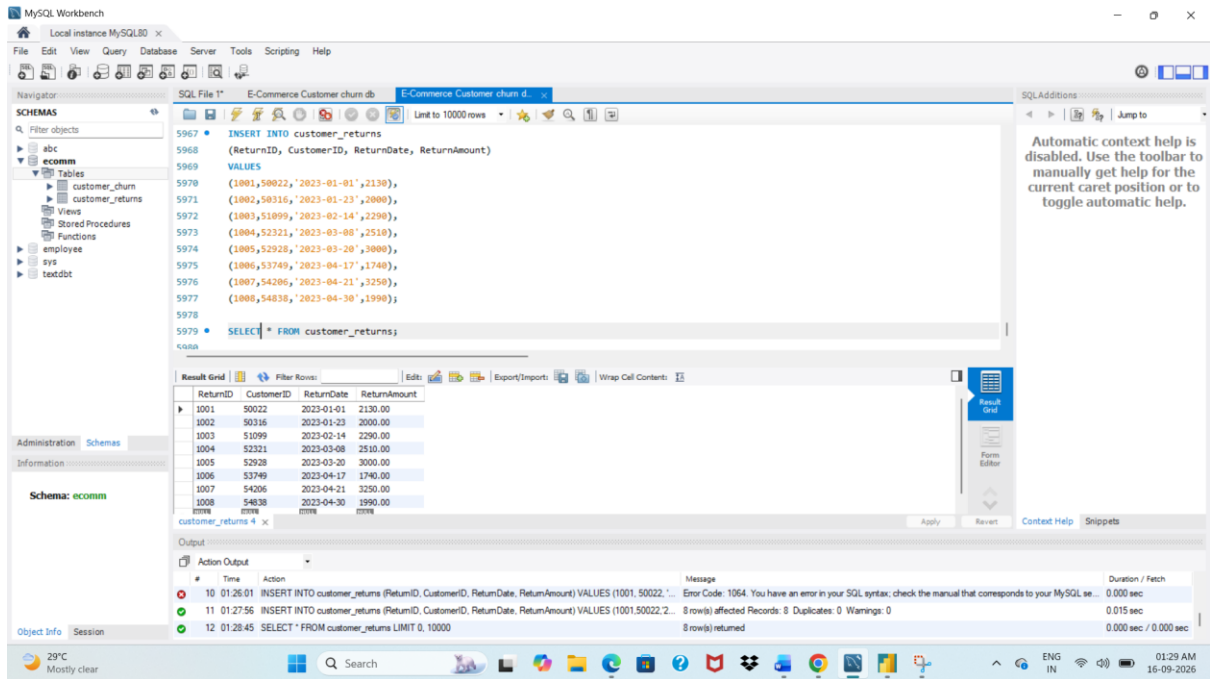
#Create a 'customer_returns' table in the 'ecommm' database and insert the data

CREATE TABLE customer_returns (
  ReturnID INT PRIMARY KEY,
  CustomerID INT,
  ReturnDate DATE,
  ReturnAmount DECIMAL(10,2));

```

The output pane at the bottom shows the execution progress:

#	Time	Action	Message	Duration / Fetch
6	22:47:49	SELECT WarehouseToHome, DistanceCategory FROM customer_churn LIMIT 0, 10000	5628 row(s) returned	0.000 sec / 0.000 sec
7	22:55:18	SELECT * FROM customer_churn WHERE MaritalStatus = 'Married' AND CityTier = 1 AND OrderCount > (SEL...	576 row(s) returned	0.016 sec / 0.000 sec
8	01:22:36	CREATE TABLE customer_returns (ReturnID INT PRIMARY KEY, CustomerID INT, ReturnDate DATE, Retu...	0 row(s) affected	0.078 sec



- B) Display the return details along with the customer details of those who have churned and have made complaints.

